2012 Bored of Studies Trial Examinations

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black or blue pen. Black pen is preferred.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- Show all necessary working in Questions 11 – 16.

Total Marks – 100

Section I Pages 1 – 7

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II Pages 8 – 23

90 marks

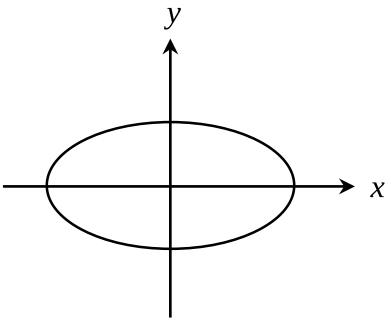
- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section.

Total marks – 10
Attempt Questions 1 – 10
All questions are of equal value

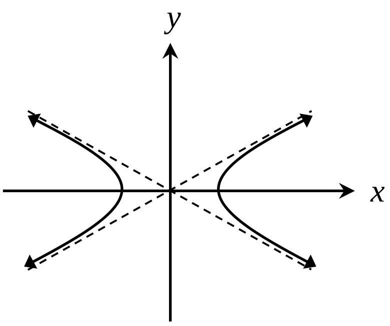
Shade your answers in the appropriate box in the Multiple Choice answer sheet provided.

1 Which of the following is the correct sketch of a conic section with eccentricity $0 < e < 1$?

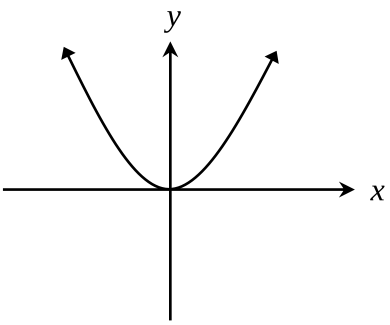
(A)



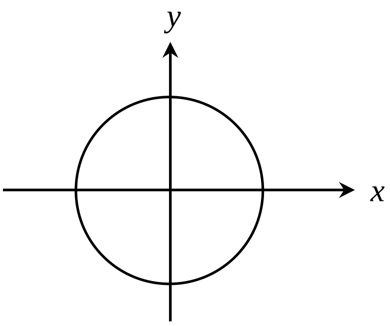
(B)



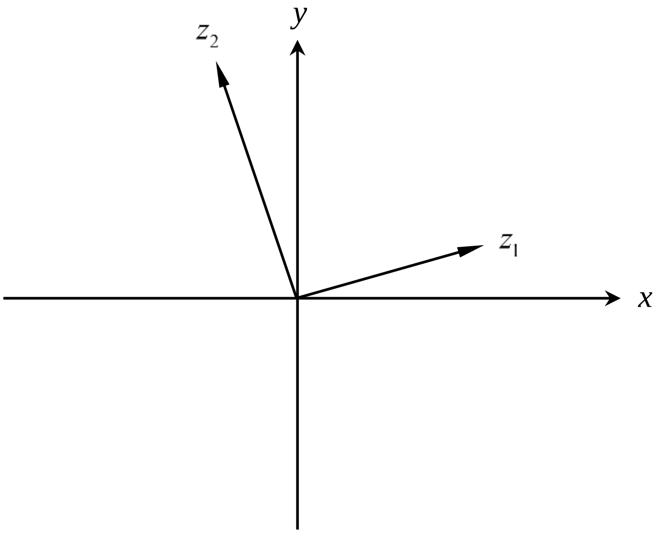
(C)



(D)

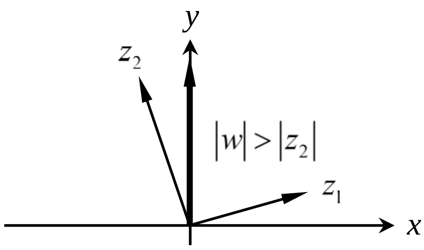


2 The following diagram shows vectors z_1 and z_2, where $|z_1| < 1$ and $|z_2| = 1$.

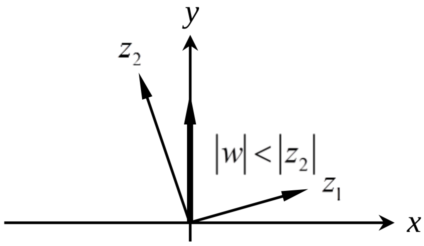


Which of the following diagrams most accurately shows the vector $w = \dfrac{z_2}{z_1}$?

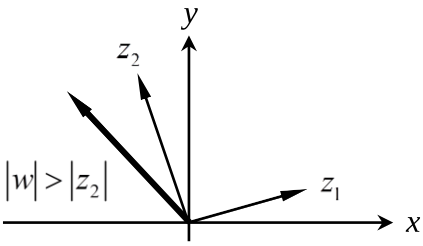
(A)



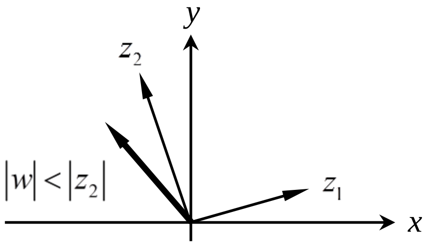
(B)



(C)



(D)



3 By letting $t = \tan\left(\frac{x}{2}\right)$, evaluate $\int \frac{dx}{1 - \sin x + \cos x}$.

(A) $\ln\left|\tan\left(\frac{x}{2}\right) - 1\right| + C$.

(B) $\ln\left|\tan\left(\frac{x}{2}\right) + 1\right| + C$.

(C) $-\ln\left|\tan\left(\frac{x}{2}\right) + 1\right| + C$.

(D) $-\ln\left|\tan\left(\frac{x}{2}\right) - 1\right| + C$.

4 Let ω be a complex root such that $\omega^n = 1$, $\omega \neq 1$.

Find the value of $\sum_{k=0}^{n}\left(\omega^k + \frac{1}{\omega^k}\right)$.

(A) 0.

(B) 1.

(C) 2.

(D) 3.

5 Which of the following statements is not necessarily true?

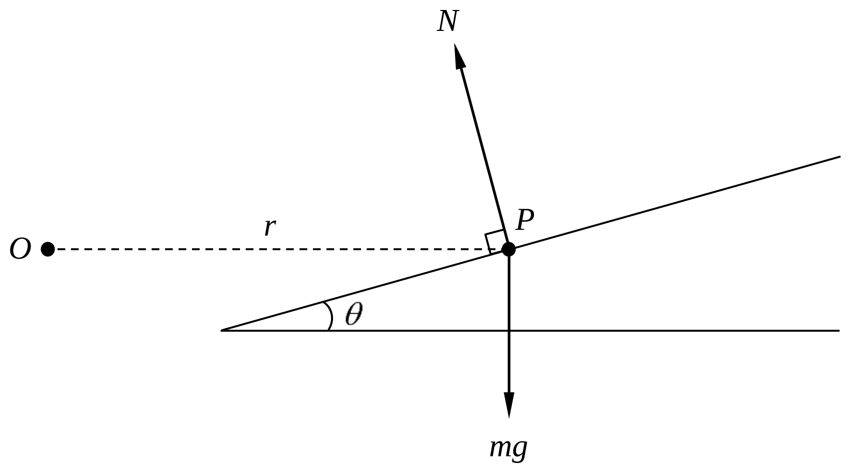
(A) $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$.

(B) If $f(x) < g(x)$ for $0 \leq x \leq a$, then $\int_0^a f(x)\,dx < \int_0^a g(x)\,dx$.

(C) If a polynomial has a root of multiplicity *n*, then the polynomial has degree n.

(D) The expression $z^n = 1$ has exactly $n-1$ non-real roots, if *n* is odd.

6 A body moves around a track with radius r, that is banked at some angle θ to the horizontal. It moves with a constant velocity $v\,\text{ms}^{-1}$. The body has some mass m and experiences a gravitational force mg, a normal reaction N to the road, and some lateral force F, which points either up or down the track, depending on the velocity.



Given that the velocity satisfies

$$0 < v < \sqrt{gr\tan\theta},$$

which of the following expressions correctly resolves the force?

(A) Horizontally: $N\sin\theta + F\cos\theta = mv^2/r$.

Vertically: $N\cos\theta - F\sin\theta = mg$.

(B) Horizontally: $N\sin\theta - F\cos\theta = mv^2/r$.

Vertically: $N\cos\theta + F\sin\theta = mg$.

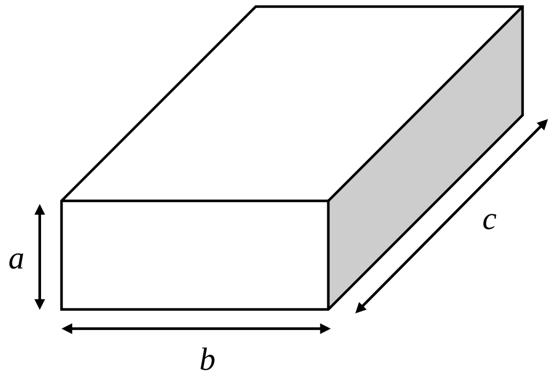
(C) Horizontally: $F\cos\theta - N\sin\theta = mv^2/r$.

Vertically: $F\sin\theta + N\cos\theta = mg$.

(D) Horizontally: $F\cos\theta + N\sin\theta = mv^2/r$.

Vertically: $F\sin\theta - N\cos\theta = mg$.

7 Consider the following rectangular prism with side lengths a, b and c such that the sum of the lengths of the edges, is equal to 4.



Find the maximum possible surface area.

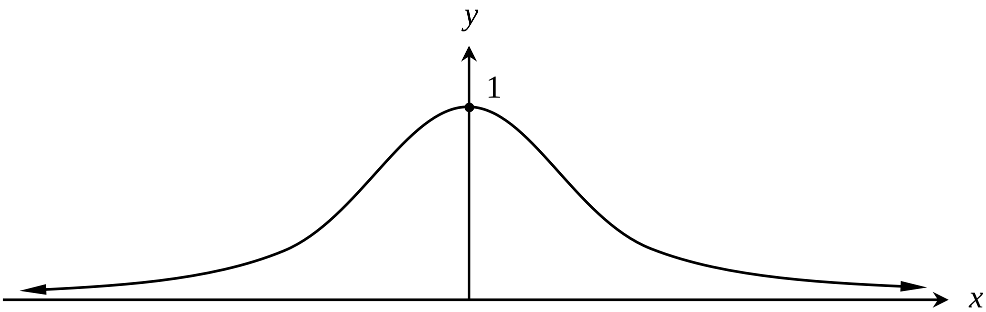
(A) $1/3$.

(B) $2/3$.

(C) 1.

(D) $4/3$.

8 Which of the following equations best describes the following curve?



(A) $y = e^{x^2}$.

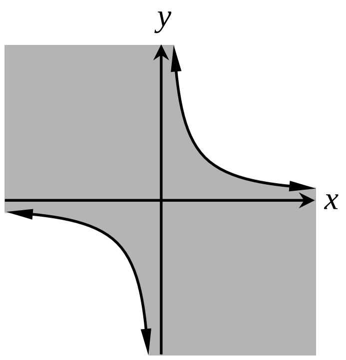
(B) $y = e^{-x^2-x}$.

(C) $y = \dfrac{1}{(x+1)^2}$.

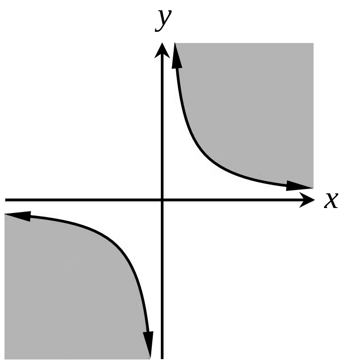
(D) $y = \dfrac{1}{x^2+1}$.

9 Which of the following is the locus of the condition $\operatorname{Im}\left(z^2\right) \leq 2$?

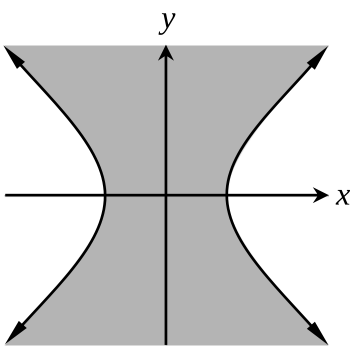
(A)



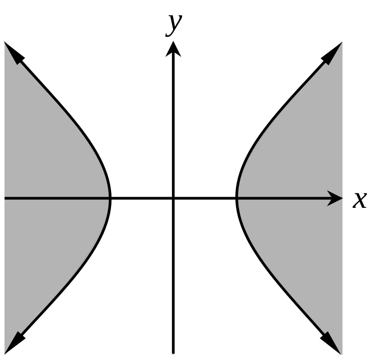
(B)



(C)



(D)



10 The polynomial $P(x)=x^6-4x^5-x^4+24x^3-5x^2-100x+125$ has a double root $2+i$.

How many of the roots of $P(x)$ are real?

(A) 0.

(B) 1.

(C) 2.

(D) 4.

Total marks – 90
Attempt Questions 11 – 16
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Evaluate $\displaystyle\int_0^1 \frac{e^{2x}-1}{e^{2x}+1}\,dx$. **2**

(b) Consider the integral $\displaystyle\int_0^{\frac{\pi}{4}} \frac{dx}{9\cos^2 x - 4\sin^2 x}$.

(i) Find functions $A(x)$ and $B(x)$ such that **3**

$$\frac{1}{9\cos^2 x - 4\sin^2 x} = \frac{A(x)}{3\cos x - 2\sin x} + \frac{B(x)}{3\cos x + 2\sin x}.$$

Hint: $\sin^2 x + \cos^2 x = 1$.

(ii) Hence evaluate the integral. **2**

(c) The region bounded by the curve $y = x$ and $y = x^n$, where $n > 1$, is rotated about the line $x = 1$.

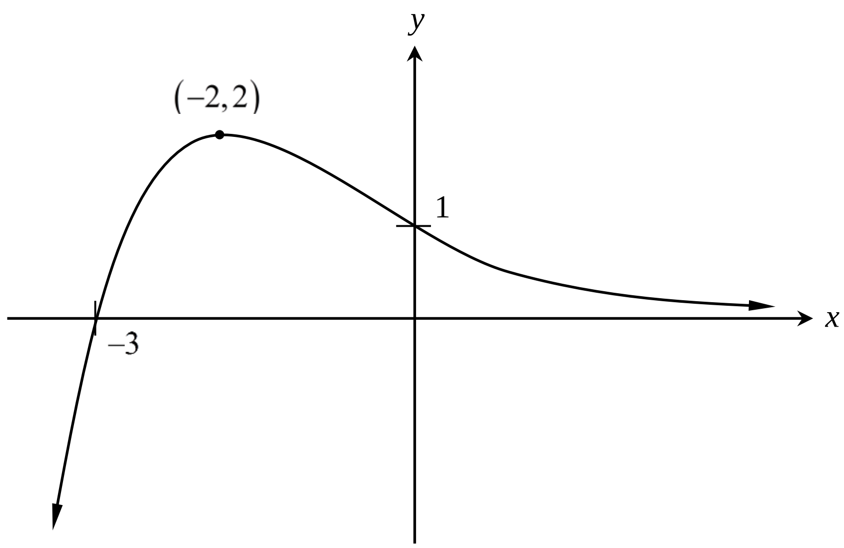
(i) Use the method of cylindrical shells to find the volume of the solid generated. **3**

(ii) Find the limiting volume of the solid as $n \to \infty$. **1**

(iii) Interpret your result geometrically. **1**

Question 11 continues on page 9

(d) The following is the graph of the function $y = f(x)$. **3**



Copy the diagram into your writing booklet.

Sketch the function $y = f\left(\frac{1}{x}\right)$, labeling all features.

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

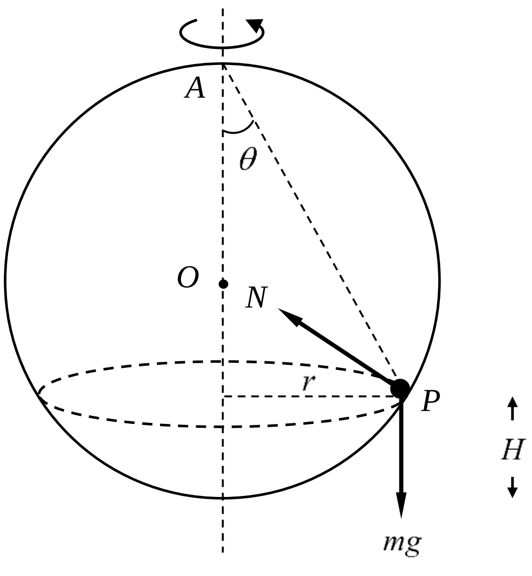
(a) Consider the curve $\sqrt{x}+\sqrt{y}=\sqrt{c}$, $c \neq 0$. A tangent is constructed in the interval $0 < x < c$, and this tangent meets the x and y axes at $A(a,0)$ and $B(0,b)$ respectively.

(i) Show that $\dfrac{dy}{dx} = -\sqrt{\dfrac{y}{x}}$. **1**

(ii) Hence prove that for any tangent in the interval $0 < x < c$, **3**

$$a+b=c\,.$$

(b) A frictionless sphere with radius R and centre O rotates about its vertical diameter with uniform angular velocity ω radians per second. A ball of mass m rolls freely inside the sphere, and experiences a gravitational force mg and a normal force N. The path of the marble describes a circle of radius $r \leq R$. Let $\angle OAP = \theta$ and let the height of the ball from the floor be H. **4**



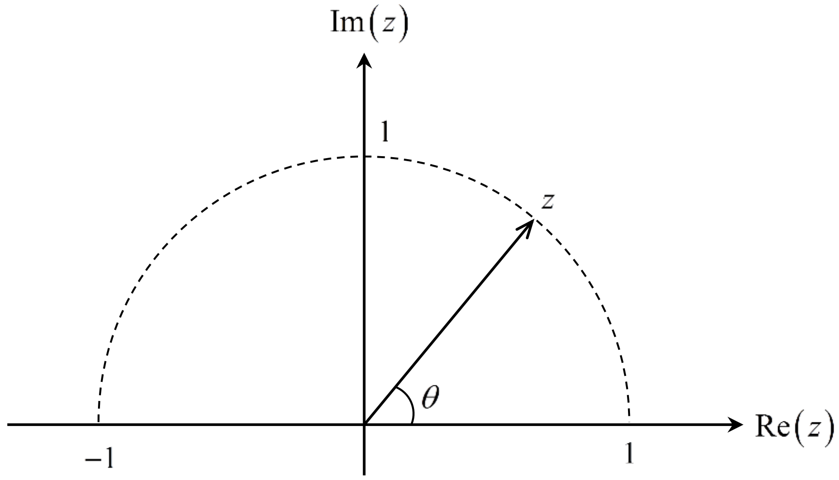
The marble rolls with some angular velocity ω such that $0 \leq h_1 \leq H \leq h_2 \leq R$. Show that

$$\sqrt{\frac{2gh_1}{r^2-h_1^2}} \leq \omega \leq \sqrt{\frac{2gh_2}{r^2-h_2^2}}\,.$$

Question 12 continues on page 11

Question 12 (continued)

(c) A complex number z lies on the upper half of a unit semi-circle.



Copy the diagram into your writing booklet.

(i) Sketch the vectors $z+1$ and $z-1$ on the same diagram. **2**

(ii) Use your diagram, or otherwise, to prove that **2**

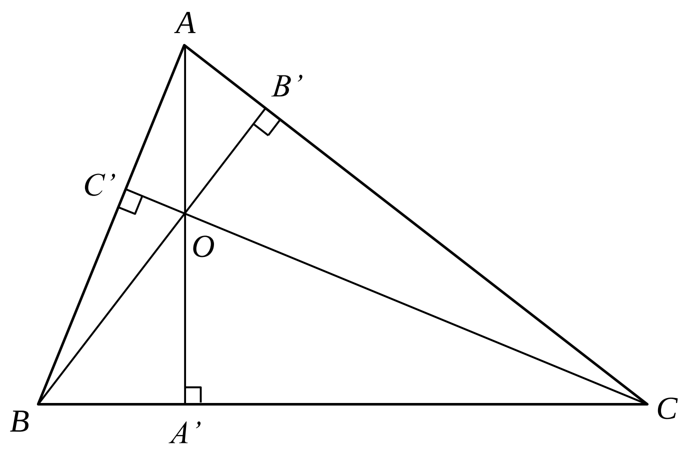
$$\operatorname{Arg}\left(\frac{z-1}{z+1}\right)=\frac{\pi}{2}.$$

(iii) Hence prove that **1**

$$\operatorname{Arg}(z)=2\tan^{-1}\left|\frac{z-1}{z+1}\right|.$$

Question 12 continues on page 12

(d) Consider an arbitrary triangle *ABC*. Three altitudes *AA'*, *BB'* and *CC'* are drawn , and they are concurrent at the point *O* **(Do NOT prove this)**.



(i) Show that **1**

$$\frac{OA'}{AA'} = \frac{|BOC|}{|ABC|},$$

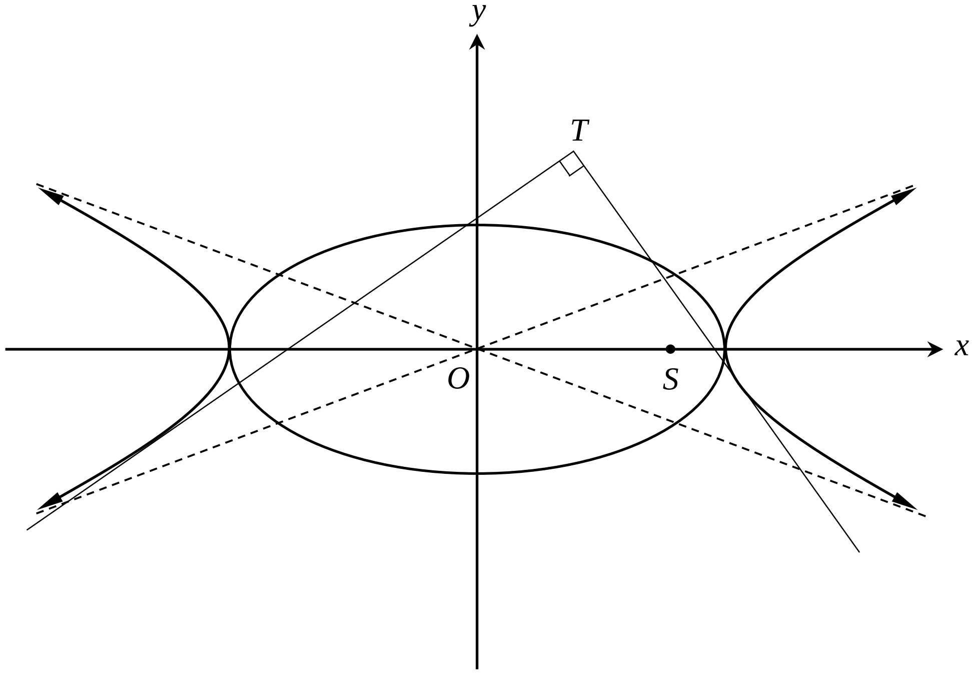
where $|XYZ|$ denotes the area of ΔXYZ .

(ii) Hence prove that $\frac{OA'}{AA'} + \frac{OB'}{BB'} + \frac{OC'}{CC'} = 1$. **1**

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) The diagram shows the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and its corresponding hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, where $a>b$, on the same set of axes. Let the positive focus of the ellipse be S. From two points on the hyperbola, mutually perpendicular tangents are drawn and intersect each other at T.



(i) Show that all lines in the form **3**

$$y = mx \pm \sqrt{a^2m^2 - b^2}\ ,$$

where m is some real constant, are tangential to the hyperbola.

(ii) Show that the locus of T is the circle $x^2+y^2=a^2-b^2$. **2**

(iii) Deduce that the triangle described by OTS is isosceles. **1**

Question 13 continues on page 14

(b) A body of mass m is projected vertically from the ground with initial velocity U. It experiences a resistance force with magnitude mkV^4, where k is a positive constant, and a gravitational acceleration g. Let its velocity at some time t be V, and also let its vertical displacement from the ground be x.

(i) Prove that **3**

$$x = \frac{1}{2\sqrt{kg}} \tan^{-1}\left(\frac{\sqrt{kg}\left(U^2 - V^2\right)}{g + kU^2V^2} \right).$$

You may assume, without proof, that $\tan^{-1} A - \tan^{-1} B = \tan^{-1}\left(\frac{A-B}{1+AB} \right)$.

(ii) Prove that the maximum height is **1**

$$H = \frac{1}{2\sqrt{kg}} \tan^{-1}\left(U^2 \sqrt{\frac{k}{g}} \right).$$

(iii) Hence find the limiting height H as $U \to \infty$. **1**

(c) A group of n people are to be arranged in a straight line. Of the n people, m particular people, where $n \geq 2m$, wish be seated together in a group. Let the number of such permutations be P_T.

All m people are now to be seated such that they are all separated by at least one person from the remaining $n - m$ people. Let the number of such permutations be P_S.

(i) Show that **3**

$$\frac{P_S}{P_T} = \frac{1}{m} \times \binom{n-m}{m-1}.$$

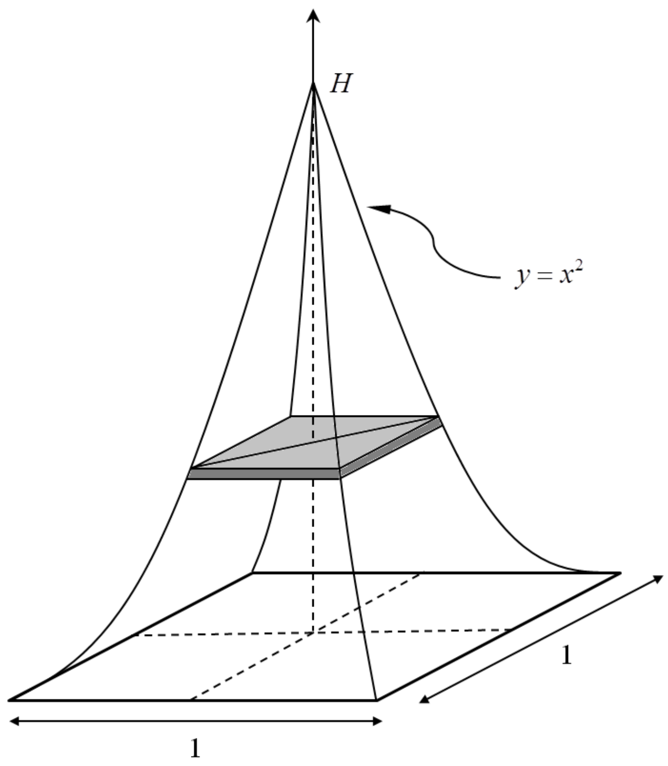
(ii) Deduce that if n is even, and m is exactly half of n, then $P_S = P_T$. **1**

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a) A pyramid-like structure with curved edges has a square base of unit length. Cross sections taken parallel to the base are squares, and the 'pyramid' eventually ends at the tip with some height H. All the curved edges follow the shape of the curve $y = x^2$, with the corners of the base being the vertex of the parabola.

Let the height, from the base, of an arbitrary slice be h.



(i) Show that the length of the diagonal of the slice is **3**

$$d = \sqrt{2}\left(1 - \sqrt{2h}\right).$$

(ii) Show that $H = \dfrac{1}{2}$. **1**

(iii) Hence, find the volume of the solid. **2**

Question 14 continues on page 16

(b) Consider the expression $z^n = 1$, which has n complex roots z_k, where $k = 1, 2, 3, ..., n$. You may assume, without loss of generality, that

$$\operatorname{Arg}(z_k) = \frac{2k\pi}{n}.$$

(i) Prove that for any two roots z_k and z_{k-1}, we have **2**

$$|z_k - z_{k-1}| = \sqrt{2 - 2\cos\frac{2\pi}{n}}.$$

(ii) The n roots of $z^n = 1$ form a regular polygon with some perimeter P_n. **1**

Explain why $P_n = n|z_k - z_{k-1}|$.

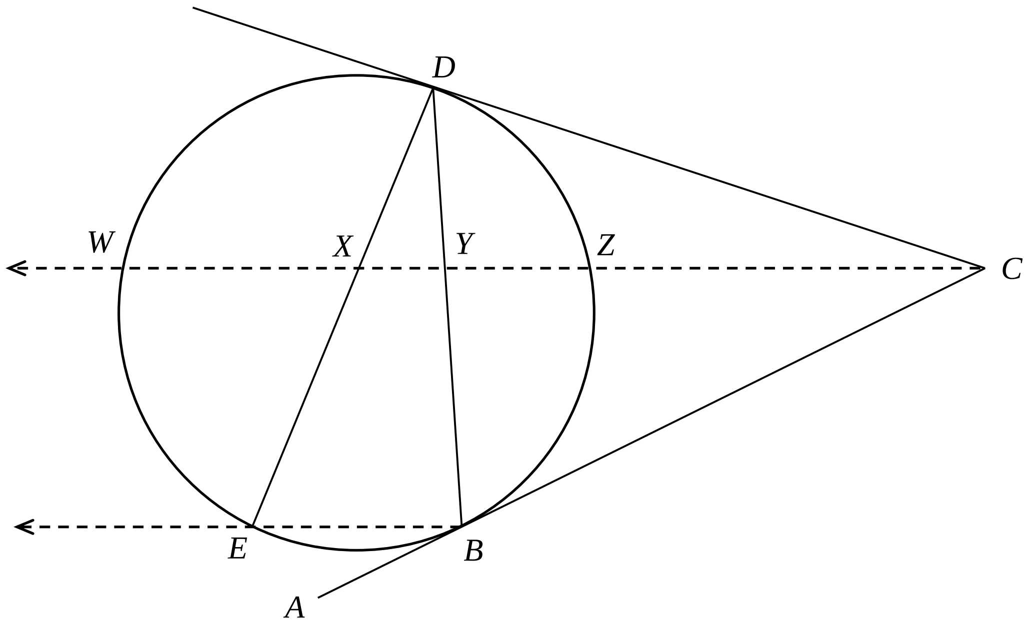
(iii) By letting $\theta = \frac{2\pi}{n}$, or otherwise, prove that **2**

$$\lim_{n \to \infty} P_n = 2\pi.$$

(iv) Explain the result in (iii) geometrically. **1**

Question 14 continues on page 17

(c) An interval AC is drawn such that it is tangential to a circle at B. From C, another tangent is constructed to meet the circle again at D. Also, a horizontal line is drawn such that it meets the circle at points Z and W shown below. A chord DB is drawn and intersects CW at Y. From point D, another chord is drawn to meet a horizontal from B at E, which lies on the circle, such that it intersects CW at X.



(i) A circle is drawn through points DXY. Explain why CD is a mutual tangent to both circles at D. **1**

(ii) Deduce that $\dfrac{XW}{XC} = \dfrac{ZY}{ZC}$. **2**

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

(a) Define the following integral, where n is a positive integer.

$$I_n = \lim_{a\to\infty}\int_0^a x^{n-1}e^{-x}\,dx.$$

You may assume that for all real $n > 0$, $\lim_{x\to\infty} x^n e^{-x} = 0$.

(i) Show that $I_1 = 1$. **1**

(ii) Show that $I_{n+1} = nI_n$. **2**

(iii) Hence prove that $I_{n+1} = n!$. **1**

(iv) Define another integral **2**

$$J_n = \lim_{b\to 0}\int_b^1 (\ln x)^n\,dx\,.$$

Show that

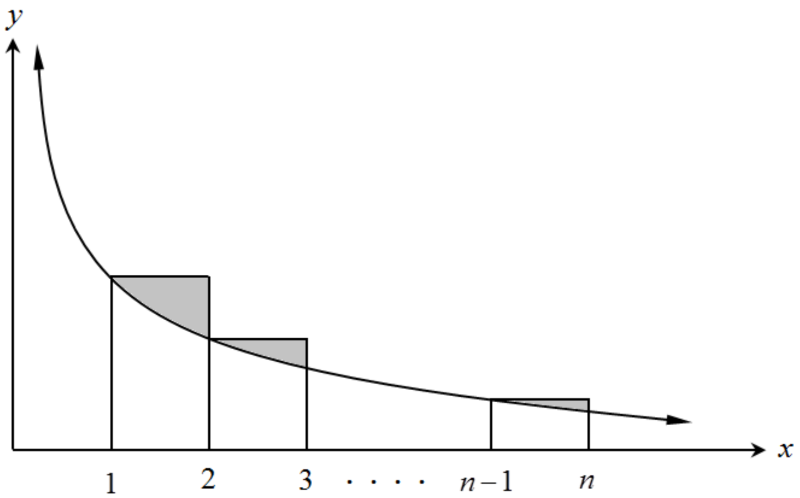
$$\frac{I_n}{J_n} = \frac{(-1)^n}{n}.$$

Question 15 continues on page 19

Question 15 (continued)

(b) Consider the function $f(x) = \frac{1}{x}$,where n is an integer. Upper bound rectangles are constructed in the domain $1 \le x \le n$, each with unit width.

Let E_n be the total excess area between the upper bound rectangles and the area under the curve, indicated by the grey regions in the diagram below.



(i) Let $H_n = \sum_{r=1}^{n} \frac{1}{r}$. Show that **2**

$$H_n - \ln n = E_n + \frac{1}{n}.$$

(ii) Given that as $n \to \infty$, E_n approaches some finite limit γ, show that **1**

$$\lim_{n \to \infty} \left(H_n - \ln n \right) = \gamma.$$

Question 15 continues on page 20

Question 15 (continued)

(iii) Define a series $S_n = \sum_{r=1}^{n} \frac{(-1)^r}{r}$. **3**

Use Mathematical Induction to prove that

$$S_{2n} = H_n - H_{2n},$$

for all positive integers *n*.

(iv) Consider I_n and J_n from part (a). Use (ii) and (iii) from part (b) to prove that **3**

$$\lim_{n\to\infty} \sum_{r=0}^{n} \frac{r!}{J_{r+1}} = \ln\left(\frac{1}{2}\right).$$

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

(a) Using the extended Triangle Inequality **(Do NOT prove this result)**, **2**

$$|z_0 + z_1 + z_2 + \dots + z_n| \le |z_0| + |z_1| + |z_2| + \dots + |z_n|$$

prove that

$$|z_0 + z_1 + z_2 + \dots + z_n| \ge |z_0| - |z_1| - |z_2| - \dots - |z_n|$$

(b) Let x be defined for all complex values.

(i) Let $R(x) = \sum_{k=0}^{n} b_k x^k$, where $n \ge 2$ and $0 < b_n \le b_{n-1} \le \dots \le b_1 \le b_0$, which has n distinct roots.

(1) Use the result in part (a) to show that **2**

$$|(1-x)R(x)| \ge b_0 - \sum_{k=1}^{n} (b_{k-1} - b_k)|x|^k - b_n |x|^{n+1}.$$

(2) Hence deduce that if $|x| < 1$, then **1**

$$|(1-x)R(x)| > 0.$$

(3) Suppose that α is a root of $R(x)$. Explain why $|\alpha| \ge 1$. **1**

Question 16 continues on page 22

(ii) Let $S(x)=\sum_{k=0}^{n} c_k x^k$, where $n \geq 2$ and $0 < c_0 \leq c_1 \leq \ldots \leq c_{n-1} \leq c_n$.

(1) Define $T(x)=\sum_{k=0}^{n} c_{n-k} x^k$. Show that if β is a root of $S(x)$, then $\dfrac{1}{\beta}$ is a root of $T(x)$. **1**

(2) Hence by considering $\left|(1-x)T(x)\right|$, or otherwise, show that **2**

$$|\beta| \leq 1.$$

Question 16 continues on page 23

Question 16 (continued)

(c) For all complex values of x, define $P(x)=\sum_{k=0}^{n} a_k x^k$, where $a_k > 0$ for $k = 0,1,2,...,n$.

For all $1 \le k \le n$, let

- A be the minimum value of $\left\{ \frac{a_0}{a_1}, \frac{a_1}{a_2}, \frac{a_2}{a_3}, ..., \frac{a_{n-1}}{a_n} \right\}$.
- B be the maximum value of $\left\{ \frac{a_0}{a_1}, \frac{a_1}{a_2}, \frac{a_2}{a_3}, ..., \frac{a_{n-1}}{a_n} \right\}$.

(i) Show that **1**

$$0 < a_n A^n \le a_{n-1} A^{n-1} \le ... \le a_2 A^2 \le a_1 A \le a_0 .$$

(ii) Let $b_k = a_k A^k$, for all $k = 0,1,2,...,n$. **2**

Prove that if ω is a root of $P(x)$, then $\frac{\omega}{A}$ is a root of $R(x)$, as defined in part (b) (i).

(iii) By letting $c_k = a_k B^k$, for all $k = 0,1,2,...,n$, show that $\frac{\omega}{B}$ is a root of $S(x)$, as defined in part (b) (ii). **2**

(iv) Hence, using (b), prove that for any root ω of $P(x)$, **1**

$$A \le |\omega| \le B .$$

End of Exam

STANDARD INTEGRALS

$$\int x^n \, dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} \, dx = \ln x, \quad x > 0$$

$$\int e^{ax} \, dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} \, dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} \, dx = \sin^{-1} \frac{x}{a}, \quad a \neq 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} \, dx = \ln\left(x + \sqrt{x^2 - a^2}\right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} \, dx = \ln\left(x + \sqrt{x^2 + a^2}\right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$